CCCCB
BBDBA
EECCB
AB

3.4 profit

1. If accounting profit is greater than zero for a firm in a competitive industry, then which of the following must be true?

(A) The firm has no fixed cost.

(B) The firm has positive economic profit.

(C) The firm's total revenue exceeds its explicit cost. ✓

(D) The firm's economic profit exceeds its opportunity cost.

(E) New firms will enter the industry in the long run.

Answer C

Correct. Accounting profit is the difference between total revenue and total explicit cost; thus, if accounting profit is greater than zero, total revenue must be greater than the total explicit cost for a firm.

2. Raheem is currently working as a financial analyst earning $\$75{,}000$ a year and is considering quitting his current job to start an art gallery. The estimated annual revenue from the art gallery is $\$175{,}000$. The annual cost of labor, advertising, and acquiring the art inventory is $\$125{,}000$. What are Raheem's accounting and economic profits if he opens the art gallery?

(A) Accounting profit is $-\$25,000$, and economic profit is $\$50,000$.

(B) Accounting profit is $\$100,000$, and economic profit is $\$50,000$.

(C) Accounting profit is $\$50,000$, and economic profit is $-\$25,000$. ✓

(D) Accounting profit is $\$100,000$, and economic profit is $-\$25,000$.

(E) Accounting profit is $\$50,000$, and economic profit is $\$100,000$.

Answer C

Correct. Accounting profit is total revenue minus explicit cost. Raheem's accounting profit is $\$175{,}000\text{-}\$125{,}000=\$50{,}000$. Economic profit is total revenue minus the sum of explicit and implicit costs. Raheem's economic profit is $\$175{,}000\text{-}(\$125{,}000+\$75{,}000)=\text{-}\$25{,}000$.

3. Which of the following is ignored when calculating accounting profit?

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(A) Total revenue
(B) Variable costs
(C) Implicit costs ✓
(D) Fixed costs
(E) Total costs

Answer C

Correct. Implicit costs are part of opportunity costs, reflecting the values of the resources — such as time, financial capital, and property — contributed by the owners of the firm. Implicit costs do not constitute out-of-pocket payments. They are not included in calculating accounting profit.

4. Which of the following statements regarding accounting profits, opportunity costs, and economic profits is true?

(A) With positive opportunity costs, a firm can never earn economic profits.
(B) Accounting profits are equal to economic profits minus opportunity costs.
(C) If accounting profits are less than opportunity costs, there will be economic losses. ✓
(D) Economic profits must always be greater than accounting profits.
(E) When economic profits are positive, accounting profits may be positive or negative.

Answer C

Correct. Total revenue minus explicit costs equals accounting profits. Subtracting opportunity costs from accounting profits yields economic profits. Therefore, if accounting profits are less than opportunity costs, there will be economic losses.

5. A firm produces 400 books and sells each book for $15. If the explicit cost of producing the books is $4,500 and the implicit cost is $1,000, the firm's economic profit is

(A) $0
(B) $500 ✓
(C) $1,000
(D) $1,500
(E) $5,000

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6. An entrepreneur has earned enough total revenue to cover her accounting costs, but economic losses are being incurred. What must be true?

(A) Her accounting costs are larger than her economic costs.

(B) Her accounting profits are less than her implicit costs. ✓

(C) Her accounting profits are greater than her economic costs.

(D) Her economic losses are less than her total fixed costs.

(E) Her implicit costs are less than her accounting costs.

Bruce is a talented writer and graphic artist who enjoys both types of work equally. Instead of earning $45,000 as a writer, Bruce now earns $25,000 in accounting profits as a graphic artist using the same computer equipment he would have used as a writer.

7. What is Bruce's economic profit from choosing to work as a graphic artist?

(A) -$45,000

(B) -$20,000 ✓

(C) $20,000

(D) $45,000

(E) $70,000

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8. Shelby is an entrepreneur who has decided to open a small advertising firm. She rents office space at a cost of $25,000 per year, she has employed an assistant at a salary of $30,000 per year, and she incurs annual utility and office supply expenses of $20,000. Her best alternative is to work elsewhere and to earn a salary of $50,000 per year. How much annual revenue must her firm receive so that Shelby earns zero economic profit?

(A) $50,000

(B) $75,000

(C) $100,000

(D) $125,000 ✓

(E) $150,000

9. Habib withdrew $100,000 from his bank account paying 5% interest to purchase equipment for his construction company. If Habib earns an accounting profit of $10,000 and he has no other opportunity costs, his economic profit will be equal to

(A) $0

(B) $5,000 ✓

(C) $15,000

(D) $105,000

(E) $115,000

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Answer B

Correct. Habib earns an accounting profit of $10,000. Subtracting the $5,000 implicit cost associated with the forgone interest income, Habib earns a positive economic profit of $5,000.

The following questions are based on the table provided.

The table provided shows a firm's average fixed cost (AFC), total variable cost (TVC), and total economic profit at different levels of output in units.

Output	AFC	TVC	Profit
50	$20		
200		$1,600	$1,000

10. If the marginal revenue of the 201st unit of output is $15 and the marginal cost of the 201st unit is $20, the total economic profit of producing 201 units will be

(A) $995 ✓

(B) $1,000

(C) $1,005

(D) $1,015

(E) $1,035

Answer A

Correct. The marginal (additional) economic profit of the 201 st unit is the difference between the marginal revenue of the 201 st unit and the marginal cost of the 201st unit ($15 − $20) = −$5. The marginal (additional) economic profit of −$5 at the 201 st unit reduces the total economic profit of producing 201 units to $995, since the total profit of producing 200 units was $1, 000.

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11. Dylan recently quit his job as a journalist for a local newspaper where he was making $80,000 per year. He opened an ice-cream shop where he earns a yearly total revenue of $250,000. He spends $150,000 each year on rent, resources, and workers he hires. Which of the following equals Dylan's accounting profit and economic profit from the ice-cream shop?

	Accounting Profit	Economic Profit
A	$250,000	$20,000
B	$230,000	$100,000
C	$150,000	$70,000
D	$100,000	$80,000
E	$100,000	$20,000

(A) A
(B) B
(C) C
(D) D
(E) E ✓

Answer E

Correct. Accounting profit is the difference between total revenue and total explicit cost, which is $\$250,000\text{-}\$150,000=\$100,000$ in this case. Economic profit is the difference between total revenue and the sum of explicit and implicit costs; in this case, economic profit is $\$250,000\text{-}(\$150,000+\$80,000)=\$20,000$.

Jamal quits a job that was paying him $\$30,000$ per year and decides to start his own business. He runs his business out of his house in a room he had been renting to his colleague for $\$12,000$ a year. Jamal withdraws the $\$20,000$ in his savings account that had been earning him a 10 percent annual interest to purchase computers and related accessories and equipment for the business. During the first year of operation, Jamal's business incurred $\$30,000$ in explicit costs and generated $\$60,000$ in total sales.

12. Jamal's economic profit is

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(A) $30,000
(B) $17,000
(C) $0
(D) − $2,000
(E) − $14,000 ✓

Answer E

Correct. Economic profit is calculated as the difference between total revenue minus both explicit and implicit costs. Given the data above, Jamal's economic profit is $(\$60,000 - \$30,000 - \$30,000 - \$12,000 - \$2,000) = -\$14,000$.

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13. Assume Nadia voluntarily leaves a job with a salary of $\$100$ per day to open and run a restaurant instead. After deducting all explicit costs from the restaurant revenues, Nadia has a gain of $\$120$. Assuming there are no additional implicit costs, which of the following statements is true?

(A) Nadia has an accounting profit of $\$20$.
(B) Nadia has an accounting profit of $\$100$.
(C) Nadia has an economic profit of $\$20$. ✓
(D) Nadia has an economic profit of $\$120$.
(E) Nadia has an economic profit of $-\$20$.

Answer C

Correct. Total revenue minus explicit costs equals accounting profits. Deducting any opportunity costs from accounting profits yields economic profits. Rational decision making should include all costs, and, thus, be based on economic profit or loss. In this example, accounting profit is stated as $\$120$. Subtracting the opportunity cost of forgone wages $(\$100)$ yields an economic profit of $\$20$. Economic profit is positive since accounting profit exceeds the opportunity cost of lost wages.

14. Kieran owns and operates his own bike shop. In the past week, a competitor offered to buy Kieran's bike shop for $100,000 and hire Kieran for $50,000 per year. Assume the annual interest rate is 6 percent, and Kieran's accounting profit from his bike shop is $60,000. Kieran's economic profit is

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(A) −$96,000
(B) −$90,000
(C) $4,000 ✓
(D) $10,000
(E) $60,000

Answer C

Correct. The correct answer is $4,000. The $4,000 equals the total revenues minus the total explicit and implicit costs. The total opportunity cost is the sum of the $50,000 salary offer plus the $6,000 interest forgone. Subtracting $56,000 from the accounting profit of $60,000 gives an economic profit of $4,000.

15. Economic profit can be calculated as accounting profit minus which of the following?

(A) Fixed costs
(B) Implicit costs ✓
(C) Marginal costs
(D) Explicit costs
(E) Total costs

16. Instead of being employed at a printing company at a salary of $25,000 per year, Sally starts her own printing firm. Rather than renting a building that she owns to someone else for $10,000 per year, she uses it as the location for her company. Her costs for workers, materials, advertising, and energy during her first year are $125,000. If the total revenue from her printing company is $155,000, her total economic profit is

(A) -$5,000 ✓
(B) $5,000
(C) $20,000
(D) $30,000
(E) $120,000

17. Ryan quit a job with a daily salary and opened a business. On a daily basis, the total revenue of the business is $200, and the explicit costs of the business are $120. If Ryan has zero economic profits, what must be the value of Ryan's implicit costs?

(A) $0
(B) $80 ✓
(C) $120
(D) $200
(E) $280

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Answer B

Correct. Total revenue minus explicit costs equals accounting profits. Subtracting opportunity cost from accounting profits yields economic profits. In this example, accounting profits are $\$200 - \$120 = \$80$. If economic profits are zero, the implicit cost (the individual's forgone wages) must be $\$80$. That is, the individual's economic profit is $\$80 - \$80 = \$0$.

18. For a perfectly competitive firm, assume the price equals a rising marginal cost at 200 units of output. At this output, average total cost is $8 and average variable cost is $5. If the price is $4, by how much can this firm reduce its losses by shutting down?

(A) $800
(B) $600
(C) $400
(D) $200 ✓
(E) $100

19. Which of the following ignores implicit costs?

(A) Normal profit
(B) Accounting profit ✓
(C) Economic profit
(D) Opportunity cost
(E) Economic cost

Answer B

Correct. Accounting profit is calculated as the difference between total revenue and total explicit cost. Thus, accounting profit does not account for implicit costs.